

STUDYHUB
CLASS 10 — MP BOARD
MATHEMATICS (STANDARD)
SAMPLE QUESTION PAPER — SET 2

Time: 3 Hours	Maximum Marks: 75
Medium: English	Set: 2

Note: This is an original StudyHub practice paper based on the MPBSE syllabus and examination pattern. It is not an official MPBSE paper.

GENERAL INSTRUCTIONS

1. All questions are compulsory.
2. Questions 1 to 5 are objective-type questions.
3. Internal choices are given in Questions 6 to 23.
4. Use of a calculator is not permitted.
5. Draw neat and clean diagrams wherever required.
6. Show necessary steps in numerical questions.

PART A — OBJECTIVE TYPE QUESTIONS

Q.1 Choose the correct alternative. [6 Marks]

(1) The HCF of 96 and 144 is:

- A. 24
- B. 36
- C. 48
- D. 72

(2) If the sum of the zeroes of a quadratic polynomial is 5 and their product is 6, the polynomial is:

- A. $x^2 + 5x + 6$
- B. $x^2 - 5x + 6$
- C. $x^2 + 5x - 6$
- D. $x^2 - 6x + 5$

(3) The pair of equations $3x + 2y = 12$ and $6x + 4y = 24$ represents:

- A. Parallel lines
- B. Coincident lines
- C. Intersecting lines
- D. Perpendicular lines

(4) The 15th term of the AP 2, 6, 10, 14, ... is:

- A. 54
- B. 58
- C. 60
- D. 62

(5) If $\sin \theta = 1/2$ and θ is acute, then θ is:

- A. 30°
- B. 45°
- C. 60°
- D. 90°

(6) The probability of getting a prime number when a die is thrown once is:

- A. $1/6$
- B. $1/3$
- C. $1/2$
- D. $2/3$

Q.2 Fill in the blanks. [6 Marks]

(1) The product of two consecutive positive integers is 132. The smaller integer is _____.

(2) The discriminant of the quadratic equation $ax^2 + bx + c = 0$ is _____.

(3) The coordinates of a point on the y-axis are of the form _____.

(4) The sum of the first n terms of an AP is denoted by _____.

(5) $\cot 45^\circ =$ _____.

(6) The diameter of a circle is 14 cm. Its radius is _____ cm.

Q.3 Write True or False. [6 Marks]

- (1) $\sqrt{11}$ is an irrational number.
- (2) The graph of a quadratic polynomial is always a straight line.
- (3) If the discriminant of a quadratic equation is zero, its roots are real and equal.
- (4) The coordinates of the midpoint of (x_1, y_1) and (x_2, y_2) are $((x_1 + x_2)/2, (y_1 + y_2)/2)$.
- (5) The value of $\tan 0^\circ$ is 1.
- (6) All circles are similar to each other.

Q.4 Match the following. [6 Marks]

Column A	Column B
(i) $\sqrt{25}$	(a) 0
(ii) Distance of $(-3, 4)$ from origin	(b) 5
(iii) $\tan 45^\circ$	(c) 1
(iv) $\sin 0^\circ$	(d) 5
(v) Area of circle	(e) πr^2
(vi) Volume of sphere	(f) $\frac{4}{3} \pi r^3$

Q.5 Answer in one word or one sentence. [6 Marks]

- (1) What is the degree of a quadratic polynomial?
- (2) Write the quadratic formula.
- (3) What is the distance between two points called when they have the same x-coordinate?
- (4) Write the formula for the area of a circle.
- (5) What is the value of $\cos 0^\circ$?
- (6) What is the probability of an event that cannot occur?

PART B — SHORT ANSWER QUESTIONS

Q.6 Find the HCF of 867 and 255 using Euclid's division algorithm. [2 Marks]

OR

Show that $3 + \sqrt{5}$ is irrational. [2 Marks]

Q.7 Find the zeroes of the polynomial $x^2 - 9x + 20$. [2 Marks]

OR

If α and β are the zeroes of $3x^2 - 7x + 2$, find $\alpha + \beta$ and $\alpha\beta$. [2 Marks]

Q.8 Solve: $x - y = 5$ and $2x + y = 10$. [2 Marks]

OR

Find the value of k for which the pair $kx + 3y = 7$ and $12x + ky = 14$ has a particular required relation between coefficients. [2 Marks]

Q.9 Find the 12th term of the AP 5, 9, 13, 17, ... [2 Marks]

OR

Find the sum of the first 15 terms of the AP 2, 5, 8, 11, ... [2 Marks]

Q.10 Find the coordinates of the point which divides the line segment joining $A(2, -3)$ and $B(8, 9)$ internally in the ratio 1:2. [2 Marks]

OR

Find the distance between $P(-2, 5)$ and $Q(4, -3)$. [2 Marks]

Q.11 Evaluate $\sin^2 30^\circ + \cos^2 60^\circ$. [2 Marks]

OR

If $\sec \theta = 5/4$, find $\tan \theta$, where θ is acute. [2 Marks]

Q.12 A tangent is drawn to a circle from a point 17 cm away from its centre. If the radius is 8 cm, find the tangent length. [2 Marks]

OR

From an external point P , two tangents PA and PB are drawn. If $PA = 11$ cm, find PB . [2 Marks]

Q.13 Find the area of a circle whose radius is 7 cm. Use $\pi = 22/7$. [2 Marks]

OR

Find the area of a semicircle of diameter 14 cm. Use $\pi = 22/7$. [2 Marks]

Q.14 Find the curved surface area of a cylinder of radius 3.5 cm and height 8 cm. Use $\pi = 22/7$. [2 Marks]

OR

Find the volume of a cuboid of dimensions $12\text{ cm} \times 8\text{ cm} \times 5\text{ cm}$. [2 Marks]

Q.15 Find the mode of 4,7,5,4,8,4,6,7,4. [2 Marks]

OR

Find the mean of 6,9,11,14,10. [2 Marks]

Q.16 A box contains 4 white and 6 black balls. One ball is drawn randomly. Find $P(\text{white})$. [2 Marks]

OR

A die is thrown once. Find $P(\text{number less than 3})$. [2 Marks]

Q.17 A pole 10 m high casts a shadow $10\sqrt{3}$ m long. Find the angle of elevation of the sun. [2 Marks]

OR

If $\cos \theta = 12/13$, find $\sin \theta$, where θ is acute. [2 Marks]

PART C — 3-MARK QUESTIONS

Q.18 Solve the quadratic equation $x^2 - 11x + 24 = 0$. [3 Marks]

OR

Find the roots of $3x^2 + 2x - 8 = 0$. [3 Marks]

Q.19 In two similar triangles ABC and PQR, $AB/PQ = 3/5$. If the area of triangle ABC is 27 cm^2 , find the area of triangle PQR. [3 Marks]

OR

Prove that the ratio of the areas of two similar triangles is equal to the square of the ratio of their corresponding sides. [3 Marks]

Q.20 The marks and number of students are: Marks 10,20,30,40,50 and students 5,8,12,9,6. Find the mean marks. [3 Marks]

OR

Find the median of 12,7,15,9,18,11,6,14,10. [3 Marks]

PART D — LONG ANSWER QUESTIONS

Q.21 A two-digit number is four times the sum of its digits. If 18 is added to the number, its digits are reversed. Find the number. [4 Marks]

OR

The difference of two numbers is 5 and their product is 84. Find the two numbers using a quadratic equation. [4 Marks]

Q.22 A toy is in the form of a cone of radius 7 cm and height 24 cm. Find: (i) slant height, (ii) curved surface area, (iii) total surface area. Use $\pi=22/7$. [4 Marks]

OR

A metallic sphere has radius 6 cm. Find: (i) surface area, (ii) volume. Use $\pi=22/7$. [4 Marks]

Q.23 From a point on the ground, the angle of elevation of the top of a tower is 60° . If the point is 15 m away from the foot of the tower, find the height of the tower. [4 Marks]

OR

A vertical tower stands on horizontal ground. From a point 30 m away from its foot, the angle of elevation of its top is 45° . Find the height of the tower. [4 Marks]

ANSWER KEY — SHORT

Q.1: C, B, B, B, A, C.

Q.2: 11; b^2-4ac ; (0,y); S_n ; 1; 7.

Q.3: True; False; True; True; False; True.

Q.4: (i)-(b), (ii)-(d), (iii)-(c), (iv)-(a), (v)-(e), (vi)-(f).

Q.5: 2; $x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$; vertical distance; πr^2 ; 1; 0.

Q.6 $HCF=51$. Q.7 irrational. Q.8 zeroes=4,5. Q.9 $\alpha+\beta=7/3$, $\alpha\beta=2/3$.

Q.10 $x=5, y=0$. Q.11 depends on stated coefficient relation. Q.12 12th term=49; sum=345.

Q.13 (4,1). Q.14 10 units. Q.15 $3/4$. Q.16 15 cm. Q.17 154 cm^2 .

Q.18 $CSA=176\text{ cm}^2$. Q.19 volume= 480 cm^3 . Q.20 mode=4. Q.21 $2/5$. Q.22 30° . Q.23 $5/13$.

Q.24 roots=3,8. Q.25 area= 75 cm^2 . Q.26 mean=31.25; median=11.

Q.27 number=36. Q.28 slant height=25 cm; $CSA=550\text{ cm}^2$; $TSA=704\text{ cm}^2$. Q.29 height= $15\sqrt{3}\text{ m}$; alternative height=30 m.

MARKS VERIFICATION

Section	Questions	Marks
Objective	Q.1-5	30
Short Answer	Q.6-17	24
3-Mark	Q.18-20	9
Long Answer	Q.21-23	12
TOTAL	23 Questions	75

Final verification: $30 + 24 + 9 + 12 = 75$ Marks.